Università degli Studi di Palermo

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To the Editors of
Estuarine, Coastal and Shelf Science

Dear Editors,

We submit for your consideration the manuscript **"Interacting effects of wave coupling and bed mobility on Lagrangian transport in a shallow vegetated lagoon"**, by Cicero Martins Jr., Giuseppe Ciraolo and Mauro De Marchis, as an original research paper.

The corresponding author is MSc. Cicero Martins Jr.

Coastal lagoon models are commonly validated against water level and then used to describe transport. This manuscript asks whether the first licenses the second. We ran a complete factorial in wave coupling, bed mobility and whether seagrass resistance is uniform or distributed from a satellite classification, giving eight configurations of a three-dimensional Delft3D Flexible Mesh model coupled online with SWAN, and scored each against three tide gauges and 35 GPS surface drifters over the same nine days.

The configurations are indistinguishable in water level. Their anomaly RMSE differs by at most four millimetres at any station, which is within measurement uncertainty, so a calibration conducted on water level alone would have no basis on which to prefer any of them. Over the same simulations their Liu–Weisberg skill spans 0.469 to 0.705 and their endpoint separation 336 to 625 m.

What separates them is an interaction rather than any single process. Neither wave coupling nor bed mobility improves Lagrangian skill on its own, and each is what makes the other work. Wave coupling on a fixed bed costs 0.035 to 0.042 in skill. Bed mobility without waves costs 0.071 to 0.075 and makes every one of the 35 drifters worse in both roughness treatments. Applied together they gain 0.119 to 0.232, and the two worst and the two best members of the ensemble are built from the same pair of treatments.

Splitting the trajectory error into a magnitude part and a direction part shows the two moving together across the design, so the treatments act on the velocity field as a whole rather than on separable parts of the error. Surface speed exceeds bed speed by 1.85 to 2.20 times at sub-metre mean depth, a vertical structure set by wave coupling, which is the quantitative case for solving in three dimensions.

We believe the work suits *Estuarine, Coastal and Shelf Science* on two grounds. It is a study of brackish water and lagoon phenomena and of coastal sedimentary processes, both within the journal's stated scope. And although the model is built for one basin, the question it answers is not local: the finding concerns what a class of validation data can and cannot certify about a model, and the magnitude-direction decomposition we introduce is applicable wherever drifter observations and a circulation model exist together. The prior study closest to our wave treatment, Carniello et al. (2011), appeared in this journal.

We also draw your attention to a limitation we report against our own interest. The configurations that best reproduce observed transport depend on a morphodynamic parameter we have not been able to constrain against a repeated bathymetric survey, which we state explicitly as a limitation rather than leaving to a reader to discover.

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The corresponding author affirms that:

- All authors have directly participated in the planning, execution, or analysis of this study;
- All authors have read and approved the version submitted;
- The contents of this manuscript have not been copyrighted or published previously;
- The contents of this manuscript are not now under consideration for publication elsewhere.

We hope the manuscript is suitable for review, and we would be glad to provide any further clarification.

Sincerely,

Cicero Martins Jr.
on behalf of all co-authors
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